http://qomics.io

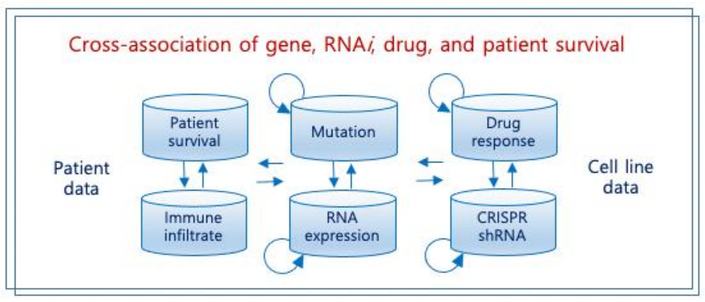
Drowning in omics data? : Q-omics, a smart software for assisting oncology and cancer research

Jieun Lee, Youngju Kim, Euna Jeong, and Sukjoon Yoon*

Center for advanced bioinformatics and systems medicine, Dept of Biological Sciences
Sookmyung Women's University, Seoul, Rep of Korea

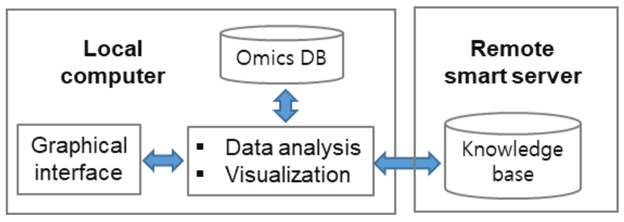
Abstract & Software workflow

Increased multi-level omics data have enabled data-driven studies on cancer drugs, targets and biomarkers. Thus, it is necessary to develop comprehensive tools for oncologists and cancer scientists to carry out extensive data mining without computational expertise. For this purpose, we have developed innovative software that enables user-driven analyses on cancer omics data, assisted by knowledge-based smart systems.



Publicly available multi-level omics data of mutations, gene/protein expression, patient survival, immune score (tumor infiltrating cells), drug screening and RNA*i* (shRNA and CRISPR) screenings on patient samples and cell lines, were integrated from the TCGA, GDSC, CCLE, NCI and DepMap databases.

Q-omics provides user-friendly interface for calculation and visualization of cross-associated data pairs retrieved from integrated datasets. The optimal selection of samples, datasets and/or other filtering options is guided by knowledge-bases of the software for the quick and easy finding of significant associations between data pairs.



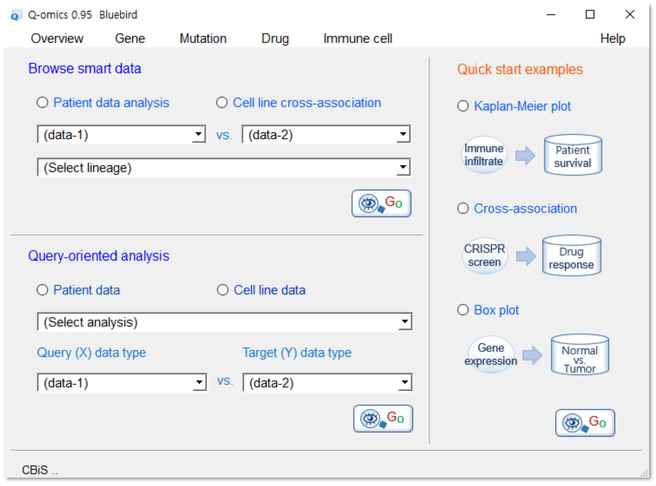
- To assist in user data mining, Q-omics interacts with a server-side knowledge base and retrieves relevant information for analysis.
- The knowledge base harbors billions of precalculated, significantly associated data pairs with related information such as sample filters and calculation options.

Smart algorithms in the knowledge base promptly select data pairs and information that is relevant to the user's query and then returns it to Q-omics. Furthermore, implemented smart algorithms prioritize significant hits based on consensus scoring methods. Consensus scoring using multiple statistical tests with varied sample (or lineage) selection, enriches noise-free, robust cross-associated pairs in the hit list. We believe that Q-omics provide simple but powerful tools for all areas of oncology and cancer research. The latest version of Q-omics software is available at http://qomics.sookmyung.ac.kr.

Q-omics

Mol. cells, 2021

Main window of Q-omics



Survival analysis

Q-omics software provides Kaplan-Meier plots and survival analysis of mutations and gene expression using TCGA datasets

1. Draw KM plots for single query or for combination of queries
2. Q-omics smart server suggests mutations, gene expression or their combinations significantly associated with patient survival.

What does the panel of Q-omics survival analysis show?

1. Graph of KM plot
2. Detailed description list of all patient risk factor (gene or mutation) data retrieved from TCGA
3. Options for KM plots such as "overall survival", "disease-free survival", stages, gender, etc.
4. Download buttons for data and plot

What does the panel of Q-omics smart search show?

1. Top-ranked list of gene/mutation combinations, lineages, stages, genders associated with patient survival for user-selected queries.
2. The smart list is derived, based on query genes or mutations that the user is interested

Cross-association analysis

Users directly calculate associations between any datasets from Q-omics panel

1. X → Y association (descriptivity) and Y → X association (predictivity) (ex. gene dataset vs. drug dataset)
2. Calculated associations are readily visualized with box plots and scatter plots

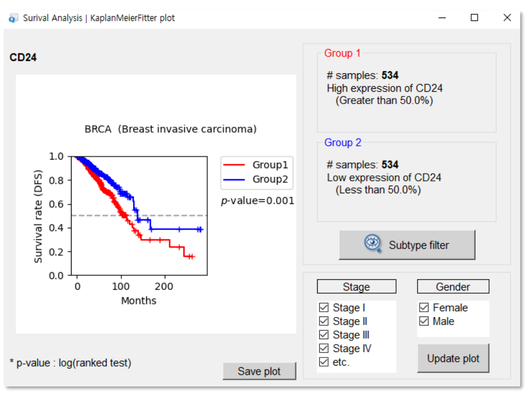
What does the panel of Q-omics cross-association show?

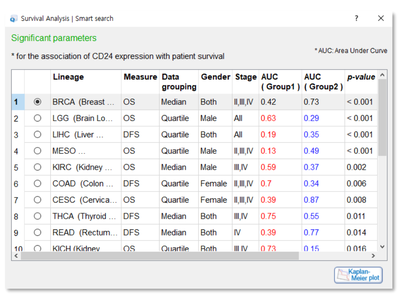
1. Graph of descriptivity and predictivity of all data points (ex. genes, RNAi or drugs)
2. Detailed description of descirptivity and predictivity in the plot
3. Options for new calculation such as p-value cutoff, data classification (median vs. quartile)
4. Box plots and scatter plots are readily generated for user-selected hits
5. Download buttons for data and plot

What does the panel of Q-omics smart search show?

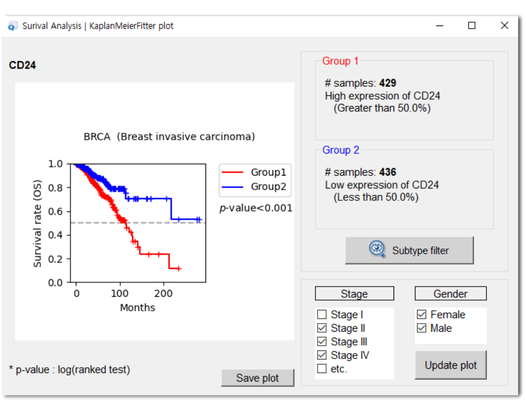
1. Top-ranked list of gene, mutation, drug, RNAi hits associated with user-selected queries.
2. The smart list is derived, based on user queries

Panel of Q-omics survival analysis and Q-omics smart search

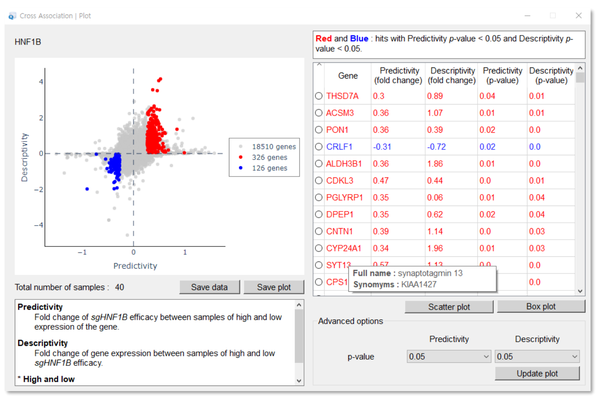


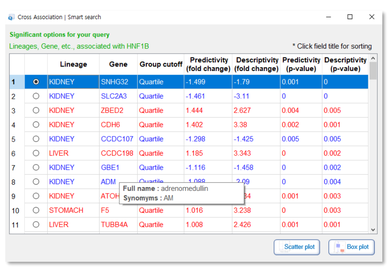
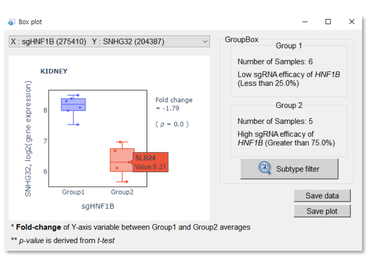
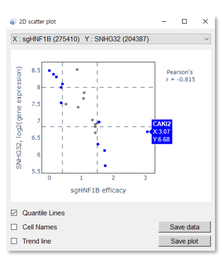


New Kaplan-Meier plot of a finding from Q-omics smart search



Panel of Q-omics cross-association analysis and smart search





Boxplots and scatter plots for smart hits